

Note 730: Minsky Papert result about symmetric polynomials to univariate ones

1: Example showing that a symmetric polynomial has a unique expansion in the elementary symmetric basis

Example 1: Express symmetric, multilinear $p \in R[x_1, \dots, x_n]$ in the elementary symmetric basis.

Let p^s be given as

$$p(x, y, z) = x + y + z - xyz$$

Observe p^s is symmetric since any permutation of its variables gives the same polynomial.

Then let $\{p_0, p_1, p_2\}$ be the set of elementary symmetric polynomials in 3 variables. That is

$$p_0(x, y, z) = 1$$

$$p_1(x, y, z) = x + y + z$$

$$p_2(x, y, z) = xy + yz + zx$$

$$p_3(x, y, z) = xyz$$

So, we can write

$$p^s = c_0 p_0 + c_1 p_1 + c_2 p_2 + c_3 p_3$$

For some real constants c_0, c_1, c_2, c_3 . In particular, take $c_0 = 0$, $c_1 = 1$, $c_3 = -1$. Then,

$$p^s(x, y, z) = p_1(x, y, z) - p_3(x, y, z)$$

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2: Showing that the polynomial from Example 1, when restricted to the Boolean domain, can be expressed as a univariate polynomial

Example 2: Write p^s in terms of a univariate polynomial.

From the previous example, $p^s(x, y, z) = p_1(x, y, z) - p_3(x, y, z)$.

We now want to show that we can write p^s as a univariate polynomial in p_1 . That is, we want to show there exists univariate polynomial $q(t)$ such that

$$p(x, y, z) = q(p_1(x, y, z))$$

In particular, observe that

$$p_1^2 = (x + y + z)^2 = 2xy + 2yz + 2xz + x + y + z = 2p_2 + p_1$$

$$p_2 = \frac{1}{2}p_1^2 - \frac{1}{2}p_1$$

Moreover, p_3 can be written as

$$3p_3 = p_1p_2 - 3p_2 = (x + y + z)(xy + yz + zx) - 3p_2$$

$$p_3 = \frac{1}{3}p_1p_2 - p_2 = \frac{1}{3}p_1 \left(\frac{1}{2}p_1^2 - \frac{1}{2}p_1 \right) - \left(\frac{1}{2}p_1^2 - \frac{1}{2}p_1 \right)$$

Where we use multilinearity of p^s . We can then simplify,

$$p_3 = \frac{1}{6}p_1^3 - \frac{1}{6}p_1^2 - \frac{1}{2}p_1^2 + \frac{1}{2}p_1$$

$$p_3 = \frac{1}{6}p_1^3 - \frac{8}{12}p_1^2 + \frac{1}{2}p_1$$

$$p_3 = \frac{1}{6}p_1^3 - \frac{2}{3}p_1^2 + \frac{1}{2}p_1$$

Indeed,

$$p^s(x, y, z) = p_1 - p_3 = \frac{1}{2}p_1^2 - \frac{1}{2}p_1 - \frac{1}{6}p_1^3 + \frac{2}{3}p_1^2 - \frac{1}{2}p_1$$

Whence, take $q(t) = \frac{1}{2}t^2 - \frac{1}{2}t - \frac{1}{6}t^3 + \frac{2}{3}t^2 - \frac{1}{2}t$ Where, for sake of sanity, we forgo simplification. Then indeed $p^s(x, y, z)$ is equal to a univariate polynomial in $x + y + z = p_0$. Namely,

$$p^s(x, y, z) = q(x + y + z)$$

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3: Example - General 2 variables symm, multi into univariate

Example 3: Let $p(x, y)$ be a real polynomial that is symmetric and multilinear that is, $p(x, y) = p(y, x)$ for all x, y and $p(x, y)$. Then there exists a real polynomial $q(t)$ such that $p(x, y) = q(x + y)$.

Proof: Let $p(x, y)$ be a symmetric polynomial. Then, by the fundamental theorem of symmetric polynomials every symmetric polynomial in n variables can be expressed uniquely as a polynomial in elementary symmetric polynomials in n variables.

In two variables the elementary symmetric polynomials are 1, $x + y$ and xy . This means that

$$p(x, y) = c_0 + c_1(x + y) + c_2xy$$

$$p(x, y) = c_0 + c_1p_1 + c_2p_2$$

Observe,

$$p_1^2 = (x + y)^2 = x^2 + y^2 + 2xy = p_1 + 2p_2$$

$$p_1^2 = 2p_2 + p_1$$

$$p_1^2 - p_1 = 2p_2$$

$$\frac{1}{2}p_1^2 - \frac{1}{2}p_1 = p_2$$

So

$$p(x, y) = c_0 + c_1 p_1 + c_2 \frac{1}{2} p_1^2 - c_2 \frac{1}{2} p_1$$

$$p(x, y) = c_0 + c_1 p_1 - c_2 \frac{1}{2} p_1 + c_2 \frac{1}{2} p_1^2$$

$$p(x, y) = c_0 + \left(c_1 - c_2 \frac{1}{2} \right) p_1 + c_2 \frac{1}{2} p_1^2$$

Take

$$q(t) = c_0 + \left(c_1 - c_2 \frac{1}{2} \right) t + c_2 \frac{1}{2} t^2$$

This gives the claim ■

4: Example - General 3 variables symm, multi into univariate

Example 4: Let $p(x, y, z)$ be a real polynomial that is symmetric multilinear. Then there exists a real polynomial $q(t)$ such that $p(x, y, z) = q(x + y + z)$

Proof:

Since p is symmetric, we can write it in terms of the elementary symmetric polynomials in 3 variables. The elementary symmetric polynomials in 3 variables are $1, x + y + z, xy + yz + xz, xyz$

This means that

$$p(x, y, z) = c_0 + c_1(x + y + z) + c_2(xy + yz + xz) + c_3(xyz)$$

For some real constants c_0, c_1, c_2, c_3 .

$$p(x, y, z) = c_0 + c_1 p_1 + c_2 p_2 + c_3 p_3$$

$$p_1^2 = (x + y + z)^2 = (x^2 + y^2 + z^2) + (2xy + 2xz + 2yz)$$

$$= 2(xy + yz + xz) + (x + y + z)$$

$$p_1^2 = 2p_2 + p_1$$

$$\boxed{p_2 = \frac{1}{2}p_1^2 - \frac{1}{2}p_1}$$

With great pain, we then compute p_1^3 (This is good counting practice, so I will do it)

$$p_1^3 = (x + y + z)^3 = \binom{3}{3}x^3 + \binom{3}{3}y^3 + \binom{3}{3}z^3 + \binom{3}{2}xyx + \binom{3}{2}xxz + \binom{3}{2}xzy + \binom{3}{2}yxx + \binom{3}{2}yyz + \binom{3}{2}yzy + \binom{3}{2}zxx + \binom{3}{2}zxy + \binom{3}{2}zyx + \binom{3}{2}zzx + \binom{3}{2}zzy + \binom{3}{2}zzz$$

$\Rightarrow$

$$p_1^3 = [x^3 + y^3 + z^3] + \binom{3}{2}[xyx + xxz + yxx + yzy + zxy + zyx + zzz]$$

$\Rightarrow$

$$p_1^3 = [x + y + z] + 6[xy + zy + xz] + 6xyz$$

$\Rightarrow$

$$p_1^3 = p_1 + 6p_2 + 6p_3$$

So,

$$p_3 = \frac{1}{6}(p_1^3 - p_1 - 6p_2)$$

Substituting in p_2 ,

$$p_3 = \frac{1}{6}\left[p_1^3 - p_1 - 6\left(\frac{1}{2}p_1^2 - \frac{1}{2}p_1\right)\right]$$

$$p_3 = \frac{1}{6}(p_1^3 - p_1 - 3p_1^2 + 3p_1)$$

$$p_3 = \frac{1}{6}p_1^3 - \frac{1}{2}p_1^2 + \frac{1}{3}p_1$$

So,

$$p(x, y, z) = c_0 + c_1p_1 + c_2p_2 + c_3p_3$$

can be written as

$$p(x, y, z) = c_0 + c_1p_1 + c_2 \left(\frac{1}{2}p_1^2 - \frac{1}{2}p_1 \right) + c_3 \left(\frac{1}{6}p_1^3 - \frac{1}{2}p_1^2 + \frac{1}{3}p_1 \right)$$

Simplifying,

$$p(x, y, z) = c_0 + \left(c_1 - \frac{c_2}{2} + \frac{c_3}{3} \right) p_1 + \left(\frac{c_2}{2} - \frac{c_3}{2} \right) p_1^2 + \frac{c_3}{6} p_1^3$$

Take

$$q(t) = c_0 + \left(c_1 - \frac{c_2}{2} + \frac{c_3}{3} \right) t + \left(\frac{c_2}{2} - \frac{c_3}{2} \right) t^2 + \frac{c_3}{6} t^3$$

5: Main theorem - A symmetric univariate polynomial can be written as a univariate

We can generalize the previous examples into the following theorem

Theorem 1: Let $p \in R[x_1, \dots, x_n]$ be a symmetric, polynomial. We claim that there is a univariate polynomial $q \in R[t]$ such that

$$\forall x \in \{0, 1\}^n: \quad p(x_1, \dots, x_n) = q(x_1 + \dots + x_n)$$

Proof:

Let $p \in R[x_1, \dots, x_n]$ be a symmetric, polynomial. Then p can be expressed as a linear combination of the symmetric elementary polynomial basis

$$p(x_1, \dots, x_n) = \sum_j^{deg(p)} c_j E_j(x_1, \dots, x_n)$$

Where c'_k s are the constants and each E_j is an elementary symmetric polynomial in n variables. We will that for each E_j there is a univariate polynomial q_j such that

$$\forall x \in \{0, 1\}^n: \quad E_j(x_1, \dots, x_n) = q_j(x_1 + \dots + x_n) \quad \text{and} \quad q(t) = \sum_{k=1}^j d_k t^k$$

We will use induction to show the claim. In particular, let $S(k)$ denote the following statement.

$$S(j): E_j(x_1, \dots, x_n) = \sum_{k=1}^j d_k E_1^k(x_1, \dots, x_n)$$

Base Case: $S(0)$

We need to show the following is true

$$S(1): E_1(x_1, \dots, x_n) = \sum_{k=1}^1 d_k q_k(x_1 + \dots + x_n), q \in R[t]$$

Trivially take $q(t) = t$, then $E_1 = q(E_1) = q(x_1 + \dots + x_n)$

This gives the claim.

Inductive Case: Suppose the $S(j-1)$ is true for some $j \geq 2$. That is, assume E_{j-1} is a degree $j-1$ elementary polynomial, such that

$$E_{j-1}(x_1, \dots, x_n) = \sum_{k=1}^{j-1} d_k E_1^k(x_1, \dots, x_n)$$

Then we want to show that $S(j)$ is true. That is we want to show

$$E_j(x_1, \dots, x_n) = \sum_{k=1}^j d_k E_1^k(x_1, \dots, x_n)$$

Consider the product E_1^j . Then note that the product of two symmetric polynomials is itself symmetric. So we can write

$$E_1^j = \sum_k^j d_k E_k$$

Split off the j th term

$$E_1^j = d_j E_j + \sum_k^{j-1} d_k E_j$$

Invoke the inductive hypothesis

$$E_1^j = d_j E_j + \sum_k^{j-1} d_k q(E_1)$$

Solve for E_j

$$E_j = \frac{E_1^j + \sum_k^{j-1} d_k q(E_1)}{d_j}$$

$$E_j = \frac{1}{d_j} \left[E_1^j + \sum_k^{j-1} d_k q_k(E_1) \right]$$

Take

$$q_j(t) = \frac{1}{d_j} \left[t^j + \sum_k^{j-1} d_k q_k(t) \right]$$

This gives the claim. ■

6 Math exchange post

While doing the examples, I realized one way we could generalize the theorem 1 (section 5) is by considering the univariate polynomial q produced by p , when restricted to boolean domain, as being represented by a change of basis matrix. So I posted this on math exchange:

How can symmetric multilinear polynomials on $\{0, 1\}^n$ can be rewritten as univariate polynomials in the sum of their variables. This produces the following matrices:

$$A_1 = [1]$$

$$A_2 = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{3} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{6} \end{bmatrix}$$

In general, I would like to know what is

$$A_n = ?$$

For context, I got these matrices by rewriting symmetric, multilinear polynomials in 1, 2, and 3 variables, respectively, into a single univariate one.

In particular,

$$p_1(x) = c_1 x$$

$$p_2(x, y) = c_1(x + y) + c_2 xy$$

$$p_3(x, y, z) = c_1(x + y + z) + c_2(xy + xz + yz) + c_3 xyz$$

Can each be written as

$$q_1(t) = c_1 t$$

$$q_2(t) = \left(c_1 - \frac{c_2}{2}\right)t + \frac{c_2}{2}t^2$$

$$q_3(t) = \left(c_1 - \frac{c_2}{2} + \frac{c_3}{3}\right)t + \left(\frac{c_2}{2} - \frac{c_3}{2}\right)t^2 + \frac{c_3}{6}t^3$$

More concretely, each polynomial q_j is equal to p_j in the following way:

$$p_1(x) = q(x)$$

$$p_2(x, y) = q(x + y)$$

$$p_3(x, y, z) = q(x + y + z)$$

Finally, the matrices A_j are the matrices that take the j -variable symmetric, multilinear polynomial p_j (when p_j is seen as a coordinate vector in the elementary symmetric basis) into the univariate q_j .

My question is, what is the name for this kind of matrix? I am unsure and would like to know if there is a general recurrence formula for A_n .

Thanks for any help!

7: Working through exponentiating multivirable polynomial

$$p_1^2 = 2p_2 + p_1$$

$$p_1^3 = p_1^2 p_1 = (2p_2 + p_1)p_1$$

$$p_1^3 = p_1 + 6p_2 + 6p_3$$

$$p_3 = \frac{1}{6} (p_1^3 - p_1 - 6p_2)$$

$$p_1^2 = (x + y)^2 = x^2 y^0 + x^1 y^1 + x^0 y^2$$

$$p_1^3 = (x + y)^3 = x^3 y^0 + x^2 y^1 + x^1 y^2 + x^0 y^3 +$$

$$p_1^3 = (x + y + z)^3 = x^3 y^0 z^0 + x^2 y^1 z^0 + x^1 y^1 z^1 + x^0 y^2 z^1 + x^0 y^1 z^2 + x^0 y^0 z^3$$

$$p_1^3 = (x + y + z + w)^3 =$$

$$p_1^3 = 3!p_3 + (p_1 + 6p_2)$$

Generally

$$p_1^n = n!p_n + r(x)$$

Where p_n is a degree n polynomial and $r(x)$ is a degree $n - 1$ polynomial.

$$p_n = \frac{p_1^n - r(x)}{n!}$$

$$p(x, y) = c_0 + c_1(x + y) + c_2xy$$

$$p_1^2 = (x + y)^2 = x^2 y^0 + 2x^1 y^1 + x^0 y^2$$